

To find a second cycle we need to show the existence of $\mathcal{C}_2$, in order to do that we need to satisfy LLL for both $\mathcal{C}_1$ and $\mathcal{C}_2$. Since neither bound is satisfied it shows that no such $\mathcal{C}_2$ exists to guarantee the existence of a second Hamiltonian cycle.

– *container*

Set $\mathcal{C}_1$ exists $\mathcal{C}_2$ such that $\mathcal{C}_1 \cup \mathcal{C}_2$